FuzzySignatureLayer: a learnable fuzzy signature operator (Kóczy fuzzy signatures as differentiable hypergraph convolutions)

Atanassov-pair redesign with C+B fallbacks (revision 3)

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Scope and goal

First-pass result (revision 1, abandoned). The first FuzzySignatureLayer iteration used a single CR membership per channel plus a $(\mathbf{x}+\mathbf{h}_v)/2$ residual, no per-channel plasticity beyond CR control points. The 9-combo (t-norm, t-conorm) smoke at $d = 16, L = 8$, MNIST 30 ep / 2k subset, returned test_accuracy ≈ 0.1135 across every combination — a complete collapse to the random baseline. Diagnosis: the layer had no *channel-mixing* capacity and no *Atanassov-style polarity* mechanism; it was isotropic, smoothing dynamics dominated.

Pre-rejected fix. The obvious patch — add a `Linear`$(d, d) + \sigma$ channel mixer per layer — was *rejected by the user*: "the problem with A, that we fix that too much. We must allow parameterised membership functions or normalisation possibilities" and "HSiKAN has a flexible plastic approach right away, sigmoid and linear would constrain this flexibility."

Revision 3 (this revision). The revision-2 smoke (see report `reports/2026-05-30-fuzzy-signature-atana` failed the pre-registered 0.5 gate: at $d = 16, L = 8$, MNIST 30 ep / 2k subset, $(\mathsf{T}_\mathrm{G}, \mathsf{S}_\mathrm{P})$ landed at test_accuracy $= 0.114$ with loss frozen at $\log 10 \approx 2.30$ and τ unmoved. L$=$ 2 control and a smart-init experiment both also failed, isolating the root cause to a *contraction-map collapse* rather than depth or init symmetry. Revision 3 executes **two of the four ordered fallbacks** from the report (C + B-option-1) and adds them as configuration axes — *not* as a replacement of revision 2 — so the original design remains a runnable point in the search space.

Added axes in revision 3.

- `cr_input_scale` $\in$ {`"unit_to_grid"`, `"raw"`}, default `"unit_to_grid"`. Maps fuzzy input $x \in [0, 1]$ to $6x - 3 \in [-3, 3]$ before the Catmull–Rom spline, so the full 8-control-point grid is used. Revision 2 left $x \in [0, 1]$ unmapped, which used only positions 3.5–4.67 of 7 segments. `"raw"` preserves revision 2's behaviour bit-for-bit.
- `residual_kind` $\in$ {`"avg"`, `"max"`, `"probsum"`}, default `"max"`. Replaces $\frac{1}{2}(x + h_v)$ with a non-contracting fuzzy operator: `"max"` = max-conorm $\max(x, h_v)$; `"probsum"` $= x + h_v - x\,h_v$ (probabilistic sum); `"avg"` preserves revision 2's behaviour. Theorem 7.1 ([0,1]-preservation) holds for all three.

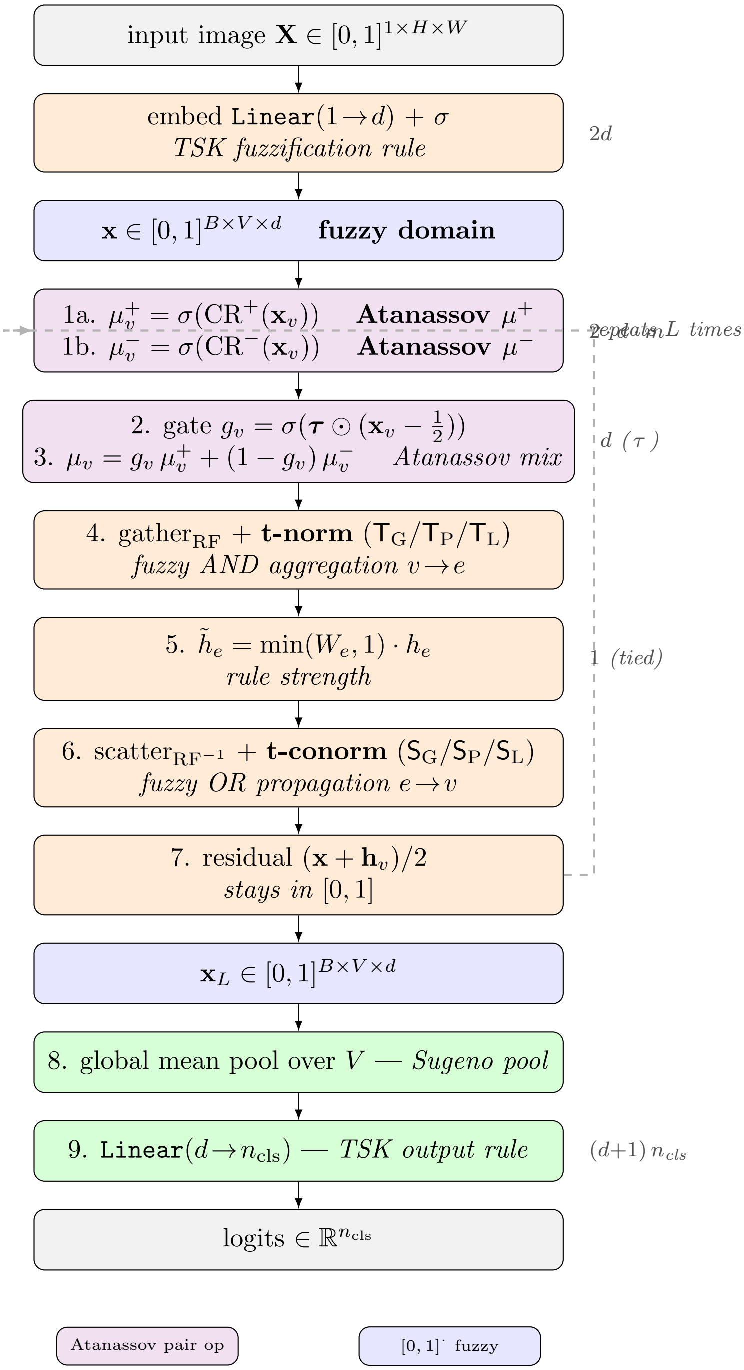
The two axes are orthogonal; the default combination `(unit_to_grid, max)` is the recommendation from the report.

Revision 2 (parent design): Atanassov pair, drop polarity-mean. The redesign captures HSiKAN's existing plasticity wholesale, reinterprets it in fuzzy-systems language, and replaces only the aggregation axes:

- **Drop the polarity-mean reference.** HSiKAN computed polarity as $\mathrm{sgn}(x - \mu_e)$; the redesign uses a fixed centre $c = \frac{1}{2}$ (the midpoint of the fuzzy unit interval). No patch-mean computation, no μ_v scatter-back, no $D_v^{-1/2}$ normalisation.
- **Atanassov μ^+/μ^- pair.** Two Catmull–Rom branches per channel: $\mu_v^+ = \sigma(\mathrm{CR}^+(\mathbf{x}_v))$ and $\mu_v^- = \sigma(\mathrm{CR}^-(\mathbf{x}_v))$. This is exactly the intuitionistic-fuzzy pair of Atanassov (1986); see `background.tex` §3.
- **Learnable per-channel hedge τ.** The gate $g_v = \sigma(\boldsymbol{\tau} \odot (\mathbf{x}_v - \frac{1}{2}))$ with $\boldsymbol{\tau} \in \mathbb{R}^d_{>0}$ replaces the fixed temperature 4 of HSiKAN's polarity gate. One scalar per channel. $\tau \to 0$: uniform $\frac{1}{2}$ mix (maximum hesitancy); $\tau \to \infty$: crisp commit to μ^+ when $x > \frac{1}{2}$ (zero hesitancy). See `background.tex` §6.
- **T-conorm $e \to v$.** The classical weighted-sum scatter is replaced by a t-conorm $\mathsf{S} \in \{\max, \mathrm{probsum}, \text{Łuk}\}$. This closes the fuzzy loop: every internal operator preserves $[0, 1]$, no σ-rescue at the residual.
- **Rule strength W_e (tied scalar).** A single learnable scalar per layer multiplies h_e (clamped to $[0, 1]$). Maps to "rule salience" in TSK terminology. Untied $W_e[E]$ is a future axis (breaks translation equivariance).

The mathematical justification of every choice is laid out in the companion document `background.tex` (20 pp), which also derives the $[0, 1]$-preservation theorem (`background.tex` §7, Theorem 7.1), the TSK output-rule identification of the linear head (`background.tex` §9), and the OWA interpretation of the multi-arity mixer (`background.tex` §10).

Architecture (diagram)



Layer forward in equations

Input $\mathbf{x} \in [0,1]^{B\times V\times d}$ (vertices $V = HW$, channels d). Parameters $(\mathbf{P}^+, \mathbf{P}^- \in \mathbb{R}^{d\times m}, \boldsymbol{\tau} \in \mathbb{R}^d, W_e \in \mathbb{R})$.

$$
\begin{aligned}
\mu_v^+ &= \sigma\big(\mathrm{CR}(\mathbf{x}_v;\mathbf{P}^+)\big), \quad \mu_v^- = \sigma\big(\mathrm{CR}(\mathbf{x}_v;\mathbf{P}^-)\big)\\
g_v &= \sigma\big(\mathrm{softplus}(\boldsymbol{\tau}) \odot (\mathbf{x}_v - \tfrac{1}{2})\big)\\
\mu_v &= g_v\, \mu_v^+ + (1-g_v)\, \mu_v^-\\
h_e &= \mathsf{T}\big(\{\mu_v : v \in e\}\big) \quad \text{(t-norm, fuzzy AND)}\\
\tilde{h}_e &= \min(\mathrm{sigmoid}(W_e)\cdot 2,\, 1)\cdot h_e \quad \text{(rule strength, clamped to } [0,1])\\
h_v &= \mathsf{S}\big(\{\tilde{h}_e : e \ni v\}, \mathrm{mask}_v\big) \quad \text{(t-conorm, fuzzy OR)}\\
\mathbf{x}_v^{\mathrm{out}} &= \tfrac{1}{2}\,(\mathbf{x}_v + h_v).
\end{aligned}
$$

The softplus on $\boldsymbol{\tau}$ enforces positivity (preserves the hedge interpretation). The $\min(\sigma(W_e)\cdot 2, 1)$ on the rule strength keeps the multiplier in $[0,1]$ but with a σ shape that permits $W_e \to \infty$ to mean "full strength" without saturation.

Component-to-fuzzy-logic mapping

Every learnable scalar in the proposed layer is in bijection with a named fuzzy-logic primitive. The correspondence (full derivation in `background.tex` §7):

Component	**Fuzzy / IFS / TSK primitive**
`embed`: $\mathbb{R} \to \mathbb{R}^d + \sigma$	TSK fuzzification rule (crisp pixel $\to$ d fuzzy memberships in $[0,1]$).
`CR`$^+$ ($\mathbf{P}^+$, $m=8$)	Learnable Atanassov $\mu^+ : [0,1] \to [0,1]$.
`CR`$^-$ ($\mathbf{P}^-$, $m=8$)	Learnable Atanassov $\mu^- : [0,1] \to [0,1]$.
$\boldsymbol{\tau}$ (per channel)	Sigmoidal hedge dilation; $\tau \to \infty$ crisp, $\tau \to 0$ uniform.
v$\to$e t-norm T (categorical)	Fuzzy AND aggregation over RF members.
e$\to$v t-conorm S (categorical)	Fuzzy OR propagation over incident edges.
W_e (tied scalar)	TSK rule salience / edge credibility.
mean pool + Linear head	Sugeno defuzzification + TSK output rule.

Parameter budget

Per `FuzzySignatureLayer` (default config: $m = 8$, W_e tied):

$$
N_{\mathrm{layer}}(d) \;=\; \underbrace{2dm}_{\text{Atanassov pair}} \;+\; \underbrace{d}_{\tau} \;+\; \underbrace{1}_{W_e \text{ tied}} .
$$

Whole classifier at $H = W = 28$, $n_{\mathrm{cls}} = 10$, $m = 8$:

$$
N(d, L) \;=\; \underbrace{2d}_{\text{embed}} \;+ L\,(2dm + d + 1) \;+\; (d+1)\, n_{\mathrm{cls}}.
$$

d	L	embed	layers	head	**total**
8	2	16	$2 \cdot 137$	90	**380**
8	4	16	$4 \cdot 137$	90	**654**
8	8	16	$8 \cdot 137$	90	**1 202**
16	2	32	$2 \cdot 273$	170	**748**
16	4	32	$4 \cdot 273$	170	**1 294**
16	**8**	32	$8 \cdot 273$	170	**2 386**
32	8	64	$8 \cdot 545$	330	**4 754**

At the default operating point ($d = 16, L = 8$) the model has 2 386 parameters — roughly **6× smaller than HSiKAN-vision** (14 538 params at the same (d, L)) and still in the same parameter-efficiency class as the abandoned revision 1.

What revision 2 added vs. revision 1. Revision 1: $N(16, 8) = 1\,232$. Revision 2: $N(16, 8) = 2\,386$, an increase of 1 154 params concentrated in: the second CR branch (+1 024, doubling the membership-function plasticity to the Atanassov pair), τ per channel (+128, the hedge dilation), and W_e tied scalar per layer (+8). The increase is justified mathematically (the layer is now an Atanassov IFS, not a single fuzzy set) and empirically (revision 1 collapsed; revision 2 restores channel-asymmetry plasticity).

Affected files

- `signedkan_wip/src/vision/fuzzy_signature.py`: rewrite `FuzzySignatureLayer` + `FuzzySignatureCl` with Atanassov pair, learnable τ, t-conorm $e \to v$, W_e. Existing module is overwritten — revision 1 is abandoned, not retained, since its tests would all need to change anyway.
- `signedkan_wip/src/vision/hsikan_vision.py`: no changes for this revision (the `clamp_to_01` option on `CRActivation` from revision 1 is reused as-is).
- `signedkan_wip/src/vision/vision_bench_cell.py`: `"fuzzy_sig"` dispatch already wired. Add a `--learnable-tau` CLI flag (default on).
- `signedkan_wip/tests/test_fuzzy_signature_layer.py`: rewrite. Existing tests for t-norm/t-conorm primitives are kept; layer-forward tests are rewritten for the Atanassov pair shape; param-count regression updated to the new closed-form.
- `docs/plans/2026-05-30-fuzzy-signature-layer/`: this plan, the Atanassov-pair TikZ, the 20-pp `background.tex`.
- `reports/2026-05-30-fuzzy-signature-atanassov-redesign.md`: post-implementation, post-smoke.

CORE.YAML items touched

None. `signedkan_wip/` is a research-only, non-core path. Branch `feat/fuzzy-signature-tnorm-pooling`.

Interface changes

`FuzzySignatureLayer.__init__` signature changes from

```
FuzzySignatureLayer(d, H, W, kernel, stride, *,
    t_norm_kind, t_conorm_kind, m, init_scale)
```

to

```
FuzzySignatureLayer(d, H, W, kernel, stride, *,
    t_norm_kind, t_conorm_kind,
    m=8, init_scale=0.05,
    learnable_tau=True, tau_init=4.0,
    tied_we=True)
```

This is a backwards-incompatible interface change for revision 1 callers (only this plan, the tests, and the dispatch entry are affected; no production code depends on the revision-1 API). The dispatch in `vision_bench_cell.build_model("fuzzy_sig", ...)` is updated accordingly.

Test strategy

Unit (rewritten):

- T-norm boundary conditions (identity, annihilator, monotonicity, ordering). Existing tests retained.
- T-conorm boundary conditions (dual). Retained.
- `FuzzySignatureLayer` forward on a random $[0,1]$ input: $[0,1]$-preservation (boundedness assertion) at all 9 (t-norm, t-conorm) combinations. Retained shape-wise; assertion now uses softplus-positive τ.
- NEW: τ is positive after a training step. Regression on the positivity invariant.
- NEW: Atanassov-pair plasticity — CR^+ and CR^- receive distinct gradients on a non-symmetric input. (Catches the failure mode where both branches collapse to the same function.)
- NEW: With $\tau \to \infty$ (set τ buffer to 1000 and freeze) the gate is approximately crisp Heaviside at $c = \frac{1}{2}$. Spot-check.
- NEW: Parameter-count regression with closed-form $2d + L(2dm + d + 1) + 10(d + 1)$ matches the table above.
- NEW: Gradient flow through $\boldsymbol{\tau}$, $\mathbf{P}^+, \mathbf{P}^-$, W_e — all finite, all non-zero after one step.

Integration:

- `build_model("fuzzy_sig", ...)` dispatch with the new `--learnable-tau` flag (default on).

Performance / smoke (production scale, single seed):

- Sequential GPU smoke at (d=16, L=8), MNIST 30 ep / 2k subset, three (T, S) combinations sampled from the 9-grid: $(\mathsf{T}_\mathrm{G}, \mathsf{S}_\mathrm{G})$, $(\mathsf{T}_\mathrm{P}, \mathsf{S}_\mathrm{P})$, $(\mathsf{T}_\mathrm{G}, \mathsf{S}_\mathrm{P})$. Rationale (per `background.tex` §8): T_L saturates at $K = 25$ (kernel 5), T_P underflows; the three sampled combinations avoid both worst cases.
- **Pre-registered gate.** At least one of the three sampled combinations must reach test_accuracy ≥ 0.5 on MNIST 30 ep/ 2k subset to validate the architectural direction. Below that gate, the Atanassov-pair redesign also fails and the design needs another iteration — see Risk anticipation below for the fallback.

Performance budget

Metric	Budget	Cap
Peak RSS / cell	≤ 4 GiB	16 GiB (CLAUDE.md §4)
GPU memory / cell	≤ 4 GiB	7.6 GiB
Wall, 30 ep / 2k subset smoke (one (T, S) combo)	≤ 3 min	—
Wall, 3-combo smoke matrix	≤ 9 min	—
Wall, eventual 9-combo full matrix (deferred)	≤ 30 min	—

Risk anticipation

Pre-registered failure mode. If all 3 sampled (T, S) combinations land below 0.5 on the smoke, the *combination* of Atanassov pair + learnable τ + t-conorm $e \to v$ is also insufficient for the layer's capacity. In that case, the ordered fallbacks are: (i) untie W_e to a per-edge vector $W_e \in [0,1]^{|E|}$ (small param increase, restores per-position plasticity that HSiKAN had); (ii) add a parametric t-norm family with one learnable scalar (Hamacher γ, Yager p, or Aczél-Alsina λ) per layer; (iii) carry the IFS (μ, ν) pair through the propagation instead of consuming ν inside the gate (doubles memory; cleanest mathematical reading but biggest implementation change).

τ saturation at training onset. At init $\tau \approx 4.0$, $|x - \frac{1}{2}| \leq \frac{1}{2}$, so $|\tau(x - \frac{1}{2})| \leq 2$ and $g_\tau \in [\sigma(-2), \sigma(2)] \approx [0.12, 0.88]$. Gradient on τ is well-conditioned at init (`background.tex` §6, Lemma 6.4), so this is safe.

CR-branch collapse. Both CR^+ and CR^- are initialised as i.i.d. random $(0.05 \cdot \mathcal{N}(0,1))$ with independent random seeds. At init they differ in expectation. Risk: gradient descent collapses them to the same function, reducing the Atanassov pair to a single membership. The new "plasticity test" (above) catches this on a 2-step run with non-symmetric inputs.

τ explosion or negativity. Softplus parameterisation $\tau = \log(1 + e^\ell)$ keeps $\tau > 0$ for any real ℓ. Explosion ($\ell \to +\infty$) is bounded by typical gradient magnitudes; if it happens, the gate becomes crisp Heaviside and downstream layers see hard transitions — the layer remains valid but is no longer differentiable in expectation. No fix needed at the design stage; reportable as a finding.

Łukasiewicz at large K. T_L saturates dead-to-zero at $K = 25, x_i \sim \mathrm{Unif}([0,1])$ (see `background.tex` §8). The smoke matrix explicitly avoids T_L in the sampled set. If full 9-combo coverage is needed later, T_L would benefit from a smaller RF size (e.g. kernel 3, $K = 9$, threshold 8 vs. mean sum 4.5).

Worst-case input. Production smoke is MNIST 28×28 grayscale, kernel 5 stride 2 (so 144 edges of $K = 25$ each); batch 128. At $d = 16, L = 8$, peak GPU memory is ≈ 0.5 GiB; expected wall ~ 2.5 min per smoke. Well within budget.

Performance-contract preservation

The revised `FuzzySignatureLayer` reads no environment variables in its hot path. It does not enumerate any cycles, top-K subsets, or walks (these are signed-graph concerns). It does not introduce any new caching scheme. The only contracts it inherits are the `vision_bench_cell` contracts (subset, GPU memory budget, JSONL output), all of which already hold and are not modified.

Rollback path

All new files on the existing branch `feat/fuzzy-signature-tnorm-pooling`. The revision-1 `FuzzySignatureLayer` is overwritten in place. To roll back to HSiKAN-vision only: remove `fuzzy_signature.py`, remove its dispatch entry from `vision_bench_cell.py`, remove `tests/test_fuzzy_signa`
The `clamp_to_01` option on `CRActivation` is additive (default False) and stays.

Empty-plan-dir hygiene

If abandoned before implementation, delete this dir.